

50 Startups CRISP-DM Feature Selection Technical Report

核心結論： 針對每一種特徵選擇演算法都已輸出獨立技術圖表。最終建議使用 R&D Spend + Marketing Spend 作為最佳化線性迴歸模型的核心特徵。

50 Startups CRISP-DM Feature Selection Technical Map

每一種演算法獨立分析：統計依據、技術圖表、模型選擇與部署結論

Records
50
No missing / no duplicate

Target
Profit
Regression problem

Best Features
R&D + Marketing
Optimized subset

Test R²
0.9168
Linear Regression

Test RMSE
\$8,206
Lower is better

CRISP-DM process

1

Business
Predict profit and identify investment drivers.

2

Data
EDA, correlation, VIF, outlier checks.

3

Preparation
Encoding, scaling, subset comparison.

4

Selection
5 algorithms validate feature value.

5

Evaluation
RMSE/R², residual, actual-vs-predicted.

6

Deployment
Serialize full pipeline by joblib.

Feature selection algorithms

Pearson
Top: R&D Spend
corr=0.973

SelectKBest
Top: R&D Spend
F=849.8

RFE
Selected: R&D + Marketing
rank=1

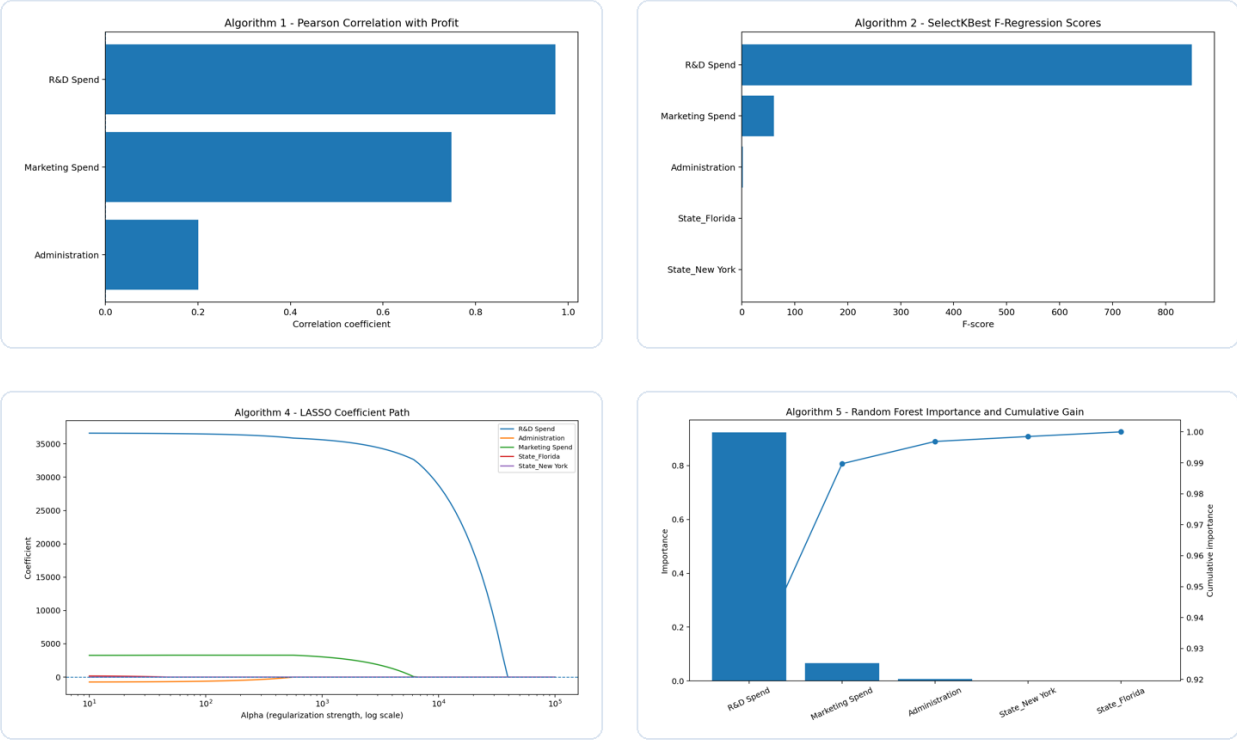
LASSO
Alpha=338
shrinks weak features

Random Forest
Top: R&D Spend
importance=92.33%

Decision matrix

Feature	Corr	VIF	Decision
R&D Spend	0.973	2.47	Keep
Marketing Spend	0.748	2.33	Keep
Administration	0.201	1.18	Drop / Optional
State	cat.	N/A	Optional / low impact

Technical charts generated for each algorithm



Final technical takeaways

1. R&D Spend is consistently selected as the strongest predictor across statistical and model-based methods.

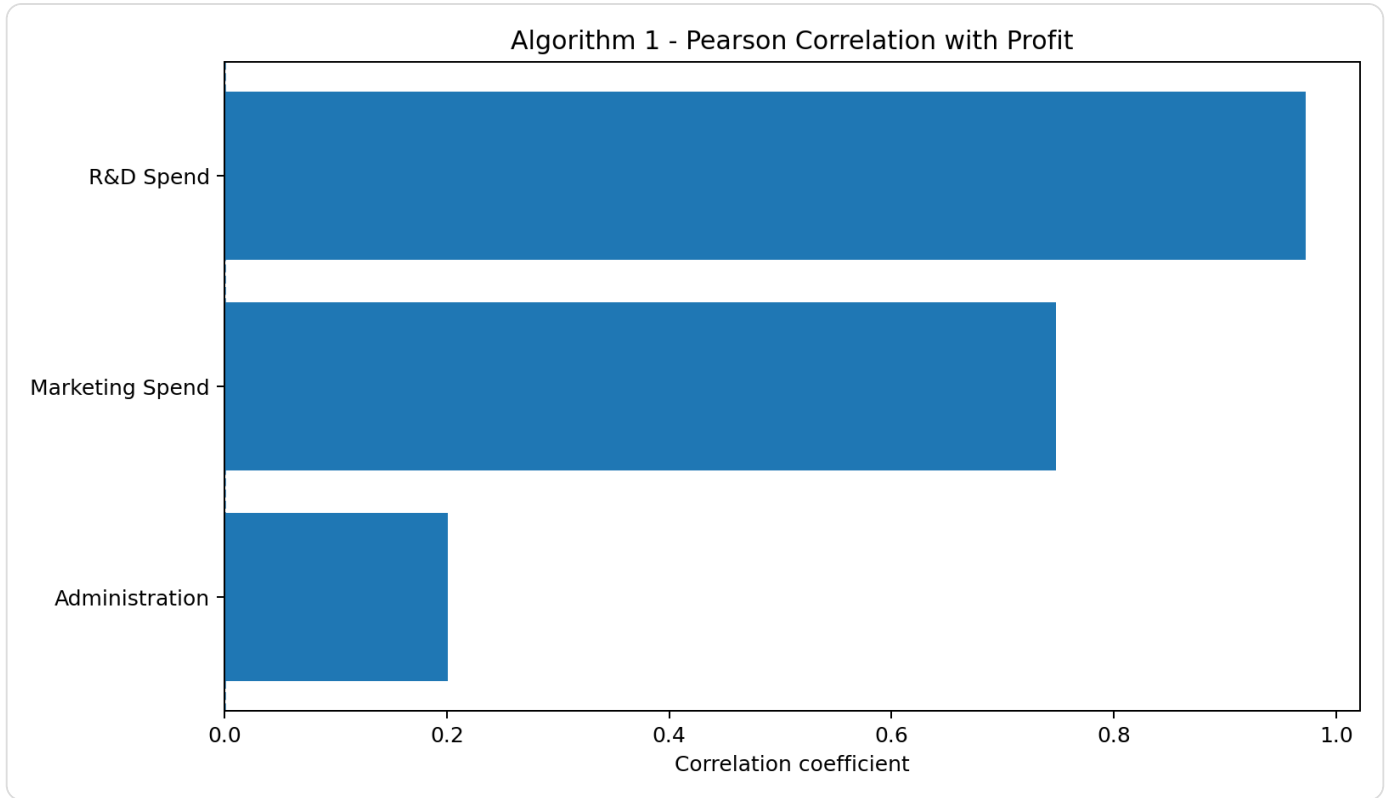
2. Marketing Spend is the second key feature and improves the optimized linear model.

3. Administration and State add limited incremental value; retain them only for baseline comparison or business interpretation.

4. Save the final preprocessing + model as one joblib Pipeline to avoid deployment mismatch.

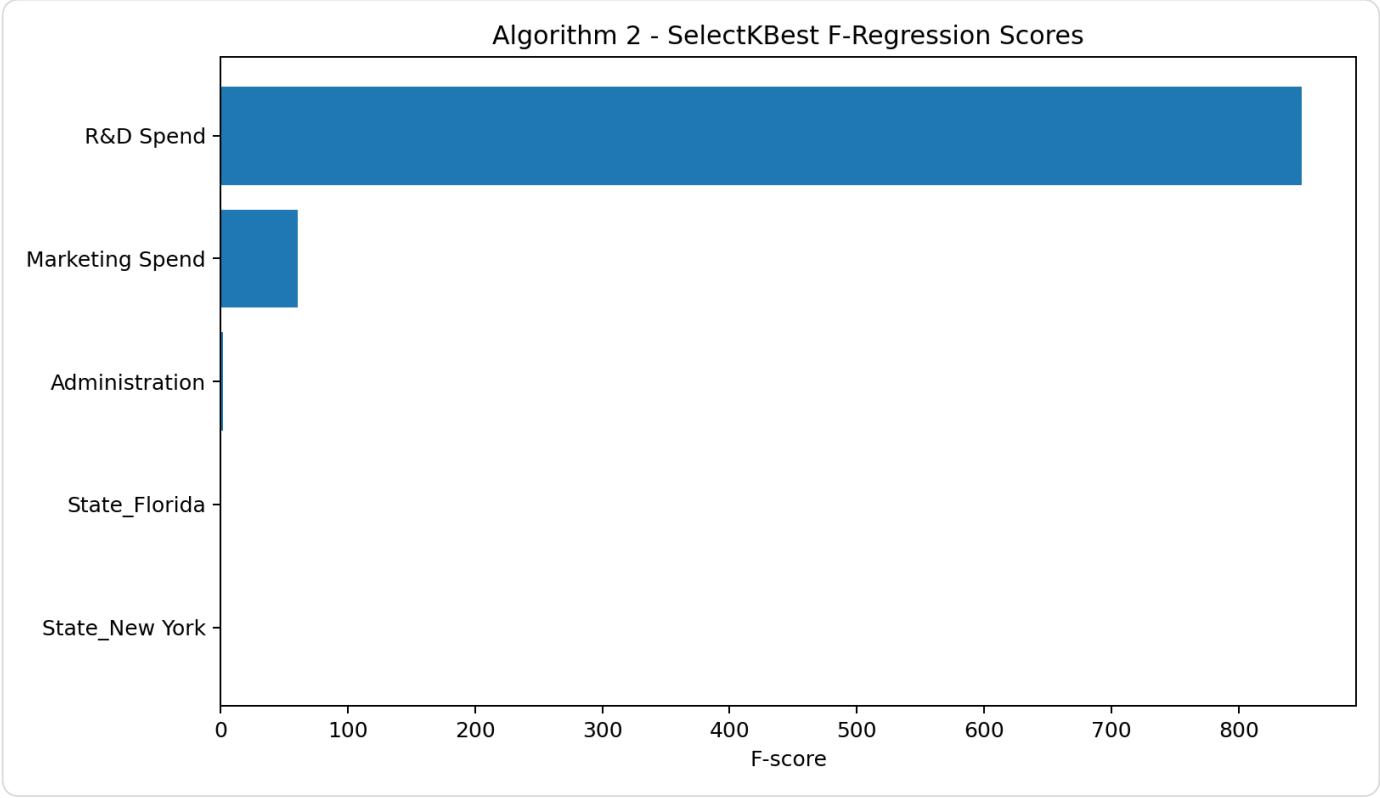
Algorithm 1 - Pearson Correlation

Feature	Correlation	AbsCorrelation
R&D Spend	0.972900	0.972900
Marketing Spend	0.747766	0.747766
Administration	0.200717	0.200717



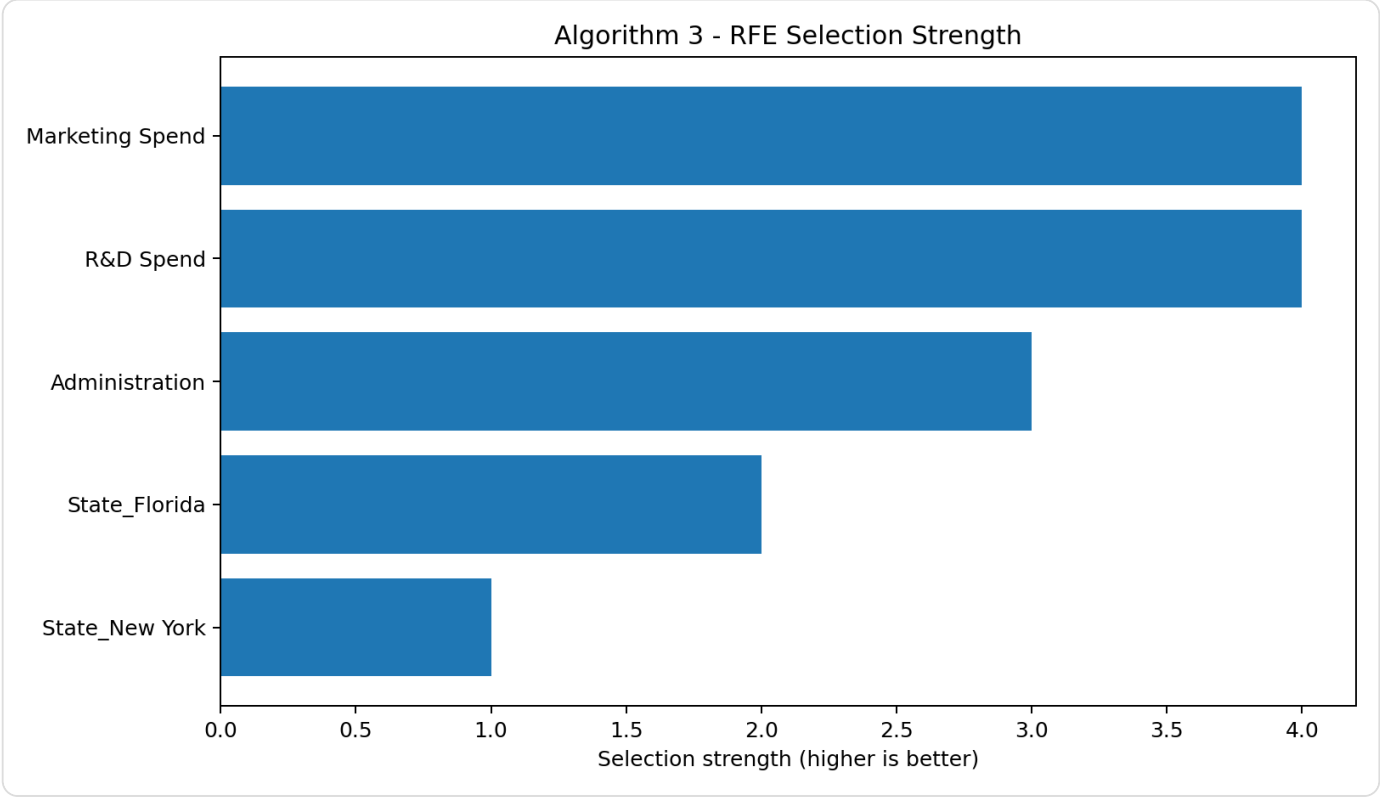
Algorithm 2 - SelectKBest F-Regression

Feature	F_score	p_value
R&D Spend	849.788904	3.500322e-32
Marketing Spend	60.881454	4.381073e-10
Administration	2.014960	1.622174e-01
State_Florida	0.657496	4.214479e-01
State_New York	0.047275	8.287963e-01



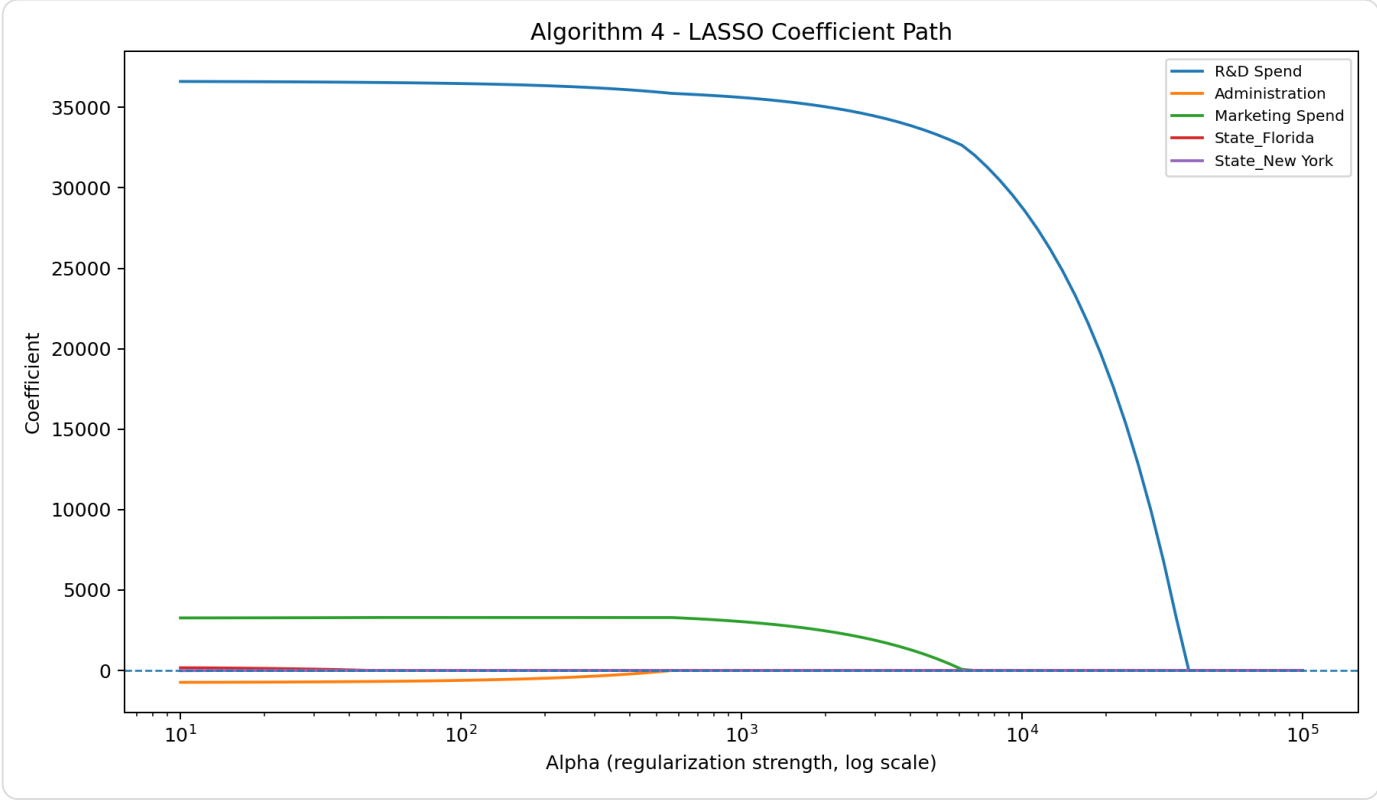
Algorithm 3 - Recursive Feature Elimination

Feature	RFE_Rank	Selected
Marketing Spend	1	True
R&D Spend	1	True
Administration	2	False
State_Florida	3	False
State_New York	4	False



Algorithm 4 - LASSO Regression

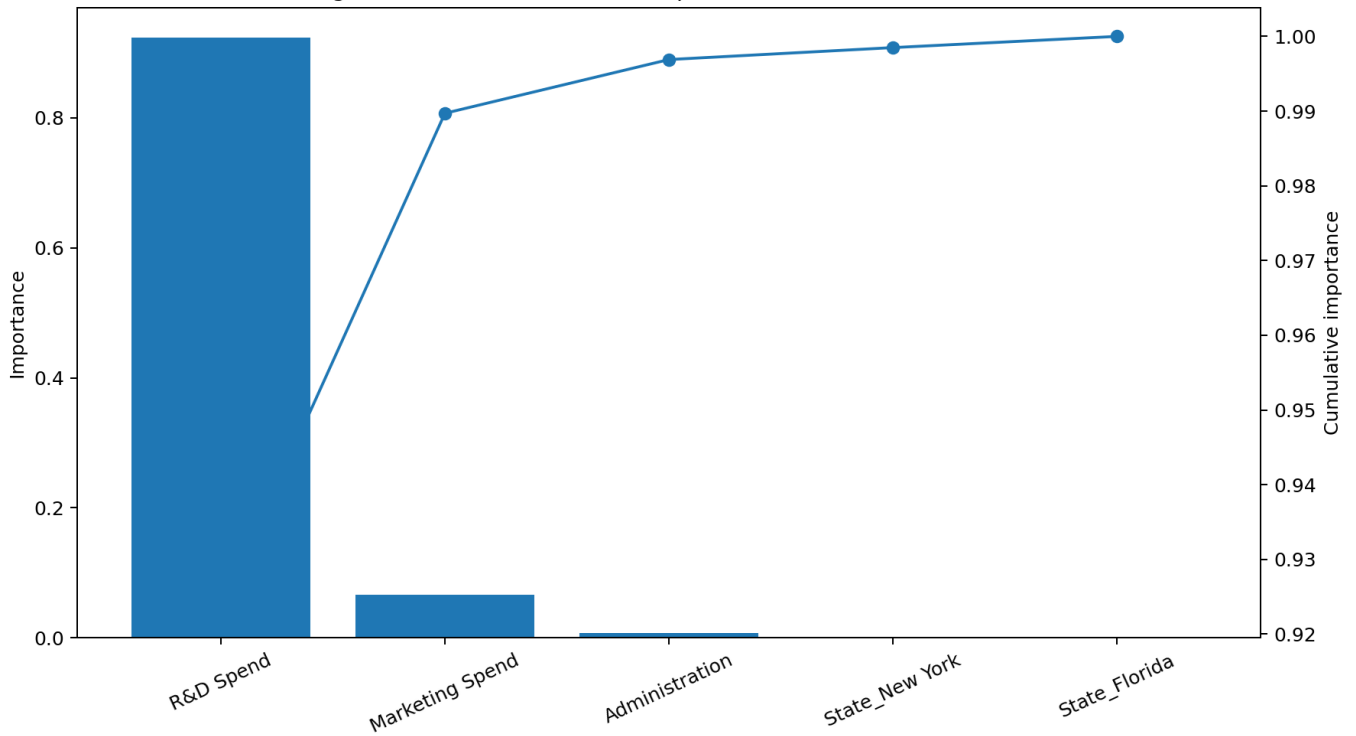
Feature	Coefficient	AbsCoefficient
R&D Spend	36173.316844	36173.316844
Marketing Spend	3290.919366	3290.919366
Administration	-300.143181	300.143181
State_Florida	0.000000	0.000000
State_New York	-0.000000	0.000000



Algorithm 5 - Random Forest Importance

Feature	Importance	Cumulative_Importance
R&D Spend	0.923340	0.923340
Marketing Spend	0.066369	0.989709
Administration	0.007192	0.996900
State_New York	0.001600	0.998500
State_Florida	0.001500	1.000000

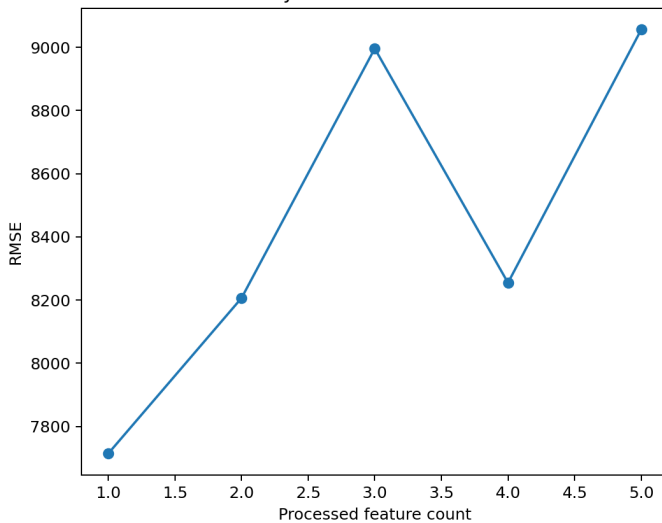
Algorithm 5 - Random Forest Importance and Cumulative Gain



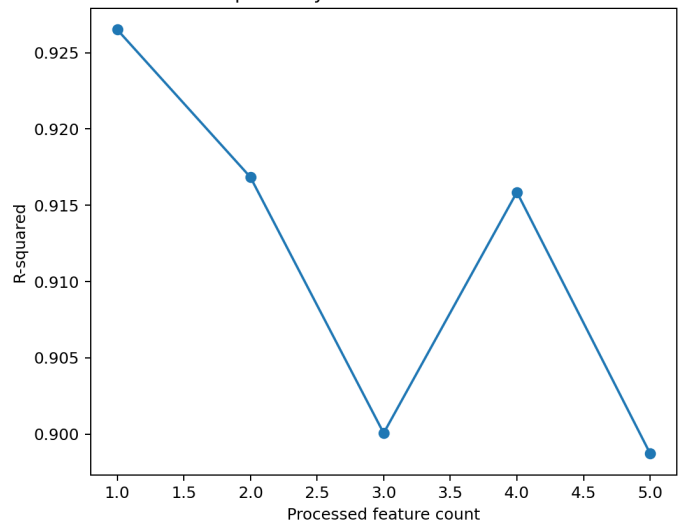
Model Subset Curves

Feature_Set	Processed_Feature_Count	MAE	RMSE	R2
1: R&D only	1	6077.363301	7714.334891	0.926511
2: R&D + Marketing	2	6469.181721	8206.328813	0.916838
3: numeric all	3	6979.152252	8995.905803	0.900065
4: R&D + Marketing + State	4	6454.510107	8254.686589	0.915855
5: all predictors	5	6961.477813	9055.957323	0.898727

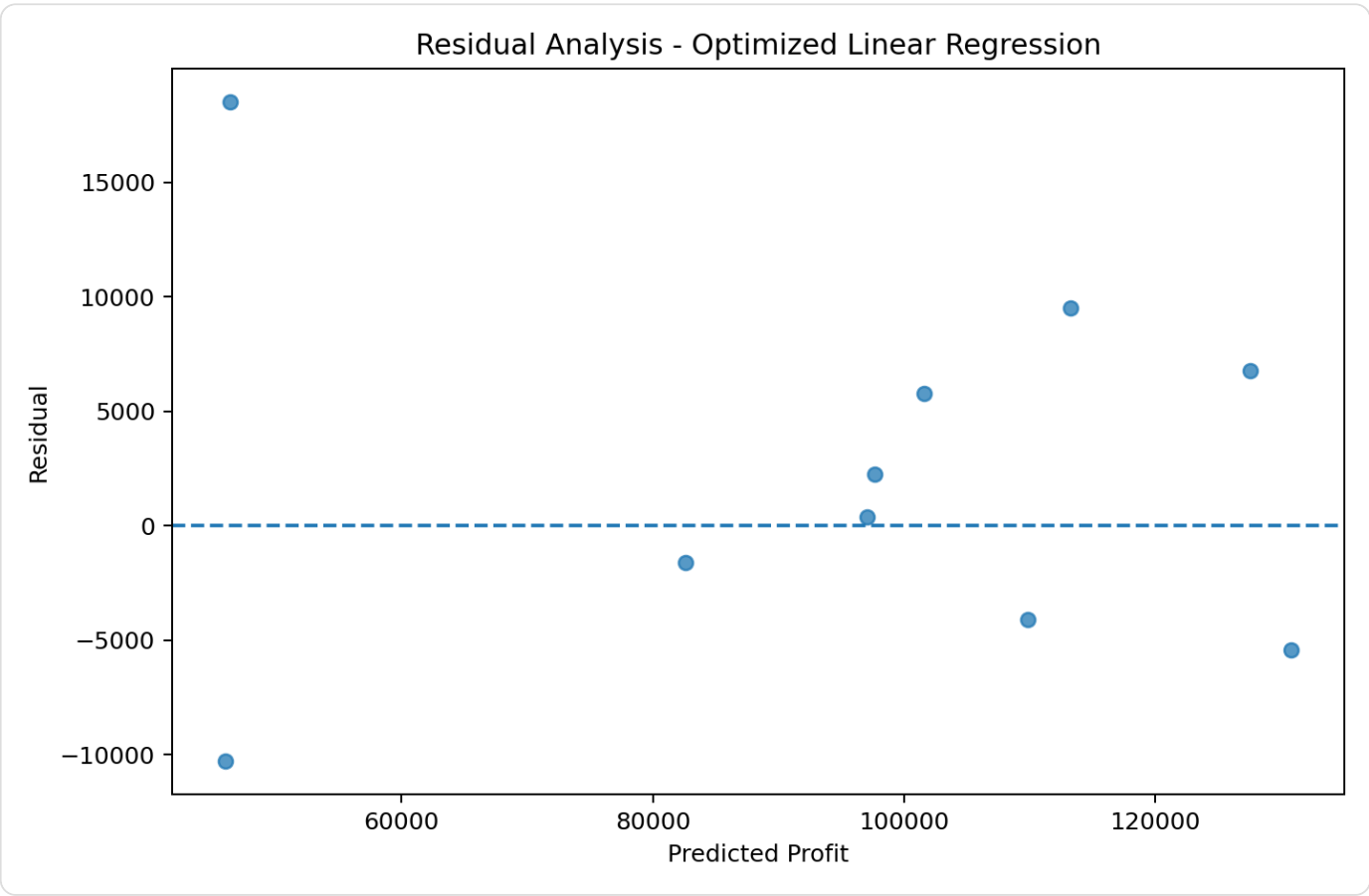
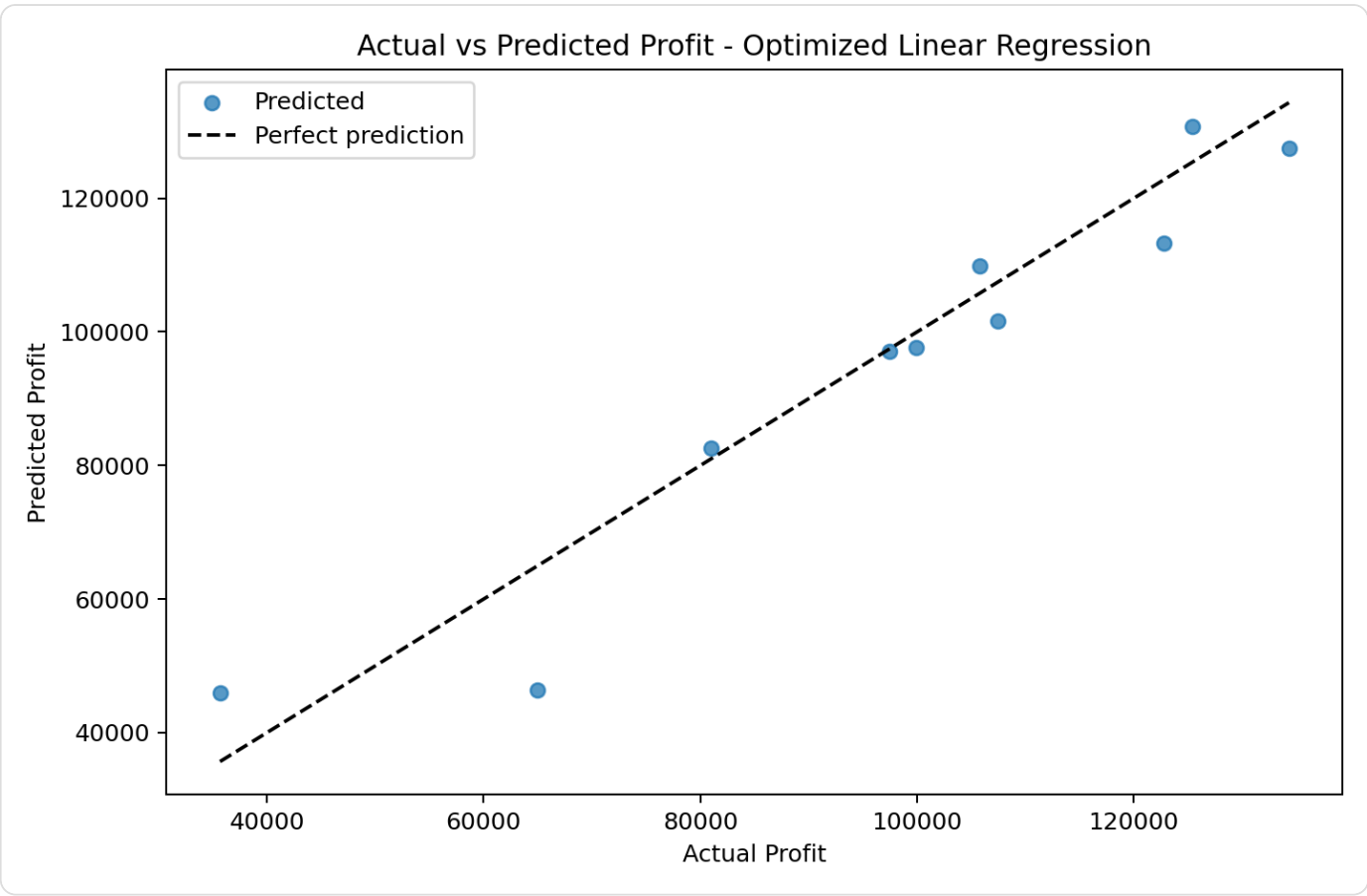
RMSE by Processed Feature Count



R-squared by Processed Feature Count



Evaluation Charts



Technical White Paper - log.md

```
# 50 Startups CRISP-DM Feature Selection Technical Report

## 1. Project Summary

This project applies the CRISP-DM methodology to the 50 Startups dataset. The goal is to predict `Profit` and identify which spending-related features provide the strongest predictive signal.

Unlike a simple regression homework, this version performs algorithm-by-algorithm feature selection analysis and produces a separate technical chart for each method.

## 2. Dataset Overview

| Item | Result |
|---|---|
| Records | 50 |
| Missing values | 0 |
| Duplicate rows | 0 |
| Target | Profit |
| Numeric predictors | R&D Spend, Administration, Marketing Spend |
| Categorical predictor | State |

## 3. Algorithm-by-Algorithm Technical Results

### Algorithm 1 - Pearson Correlation

Purpose: measure direct linear relationship between each numeric feature and `Profit`.

| Feature | Correlation |
|:-----|:-----|
| R&D Spend | 0.9729 |
| Marketing Spend | 0.747766 |
| Administration | 0.200717 |

Technical interpretation:

- `R&D Spend` is the strongest direct predictor.
- `Marketing Spend` is useful but weaker than R&D.
- `Administration` has weak direct predictive value.

Chart: `algorithm_charts/01_pearson_correlation.png`

### Algorithm 2 - SelectKBest F-Regression

Purpose: rank features by statistical relationship with the regression target.

| Feature | F_score | p_value |
|:-----|:-----|:-----|
| R&D Spend | 849.789 | 3.50032e-32 |
| Marketing Spend | 60.8815 | 4.38107e-10 |
| Administration | 2.01496 | 0.162217 |
| State_Florida | 0.657496 | 0.421448 |
| State_New York | 0.047275 | 0.828796 |

Technical interpretation:

- The F-score confirms that expenditure-related variables are more important than location features.
- The score gap supports using a compact feature subset.

Chart: `algorithm_charts/02_selectkbest_f_scores.png`

### Algorithm 3 - Recursive Feature Elimination (RFE)

Purpose: recursively remove weak features based on a Linear Regression estimator.

| Feature | RFE_Rank | Selected |
|:-----|:-----|:-----|
| Marketing Spend | 1 | True |
| R&D Spend | 1 | True |
| Administration | 2 | False |
| State_Florida | 3 | False |
| State_New York | 4 | False |

Technical interpretation:

- RFE evaluates features from a model-driven perspective.
- Features ranked 1 are the strongest candidates for the optimized model.
```

Chart: `algorithm\_charts/03\_rfe\_selection\_strength.png`

Algorithm 4 - LASSO Regression

Purpose: use L1 regularization to shrink weak coefficients toward zero.

Selected alpha from cross-validation:

```
```text
```

```
alpha = 337.6347
```
```

| Feature | Coefficient | AbsCoefficient |
|-----------------|-------------|----------------|
| R&D Spend | 36173.3 | 36173.3 |
| Marketing Spend | 3290.92 | 3290.92 |
| Administration | -300.143 | 300.143 |
| State_Florida | 0 | 0 |
| State_New York | -0 | 0 |

Technical interpretation:

- LASSO provides embedded feature selection.
- Weak or unstable predictors receive smaller coefficients.
- This is useful when expanding the dataset to higher-dimensional inputs.

Chart: `algorithm\_charts/04\_lasso\_coefficient\_path.png`

Algorithm 5 - Random Forest Feature Importance

Purpose: estimate nonlinear feature contribution using tree-based ensemble learning.

| Feature | Importance | Cumulative_Importance |
|-----------------|------------|-----------------------|
| R&D Spend | 0.92334 | 0.92334 |
| Marketing Spend | 0.0663687 | 0.989709 |
| Administration | 0.0071917 | 0.9969 |
| State_New York | 0.0016 | 0.9985 |
| State_Florida | 0.00149974 | 1 |

Technical interpretation:

- Random Forest confirms that `R&D Spend` dominates prediction.
- Marketing adds secondary value.
- State and Administration appear low-impact relative to spending variables.

Chart: `algorithm\_charts/05\_random\_forest\_importance.png`

4. Model Subset Curve

The project also compares models with different numbers of processed features.

| Feature_Set | Processed_Feature_Count | MAE | RMSE | R2 |
|----------------------------|-------------------------|---------|---------|----------|
| 1: R&D only | 1 | 6077.36 | 7714.33 | 0.926511 |
| 2: R&D + Marketing | 2 | 6469.18 | 8206.33 | 0.916838 |
| 3: numeric all | 3 | 6979.15 | 8995.91 | 0.900065 |
| 4: R&D + Marketing + State | 4 | 6454.51 | 8254.69 | 0.915855 |
| 5: all predictors | 5 | 6961.48 | 9055.96 | 0.898727 |

Technical interpretation:

- Adding more features does not automatically improve performance.
- The compact model with `R&D Spend + Marketing Spend` is recommended because it is simpler and competitive.
- The full model can remain as a baseline, but the optimized model is easier to explain and deploy.

Chart: `algorithm\_charts/06\_model\_subset\_curves.png`

5. Model Evaluation

Optimized model:

```
```text
```

```
Profit = f(R&D Spend, Marketing Spend)
```
```

| Metric | Value |
|--------------------------|---------|
| MAE | 6469.18 |
| RMSE | 8206.33 |
| R-squared | 0.9168 |
| 5-fold CV R-squared mean | 0.9389 |

Residual analysis:

- Residuals are generally centered around zero.
- No strong curve pattern is visible.
- Lower-profit companies still show larger prediction errors.

Charts:

- `algorithm\_charts/07\_actual\_vs\_predicted.png`
- `algorithm\_charts/08\_residual\_analysis.png`

6. Required Program and Report Modifications

6.1 Program changes

1. Add all five feature selection methods:

- Pearson Correlation
- SelectKBest F-Regression
- RFE
- LASSO
- Random Forest Importance

2. Save all charts as independent image files.

3. Compare baseline and optimized models separately.

4. Save the final model as a complete pipeline:

```
```text
startup_profit_pipeline.joblib
```
```

5. Add 5-fold cross-validation because the dataset has only 50 records.

6.2 Report changes

1. Do not only say "Administration was removed because of weak correlation."
2. Support the decision using multiple algorithms.
3. Explain that 'State' is evaluated but has limited incremental value.
4. Explain why the 2-feature optimized model is preferred.
5. Include residual and actual-vs-predicted plots.
6. Present all charts in a one-page infographic.

7. Final Technical Conclusion

The final technical conclusion is that startup profitability is primarily explained by R&D investment. Marketing Spend contributes secondary predictive value. Administration and State are not completely meaningless, but they add limited incremental value after the main spending variables are included.

The recommended optimized pipeline is:

```
```text
Input: R&D Spend + Marketing Spend
Preprocessing: StandardScaler
Model: Linear Regression
Output: Predicted Profit
Deployment artifact: startup_profit_pipeline.joblib
```
```

This design is simpler, explainable, and ready for Streamlit or API deployment.